

Math 330 Exam 1 Key Fall 2023

1. $\mathbb{Z}_{10}$, $U(11)$, D_5

2. $axb = c$ means $x = a^{-1}cb^{-1}$

3. The order of a group G is the number of elements in G .

The order of an element x is the smallest positive integer n such that $x^n = e$. If there is no such n we say the order of x is infinite

4. (a) 3^{-1} in $\mathbb{Z}_{20}$ is 17 (b) 3^{-1} in $U(20)$ is 7

(c) H^{-1} in D_{20} is H (d) $(a^7)^{-1}$ in $\langle a \rangle$ if $|a| = 20$ is a^{13}

5. (a) subgroup of order 2 is $\{1, 3\}$

(b) $C(3) = \{1, 2, 3, 4, 5, 6, 7, 8\}$

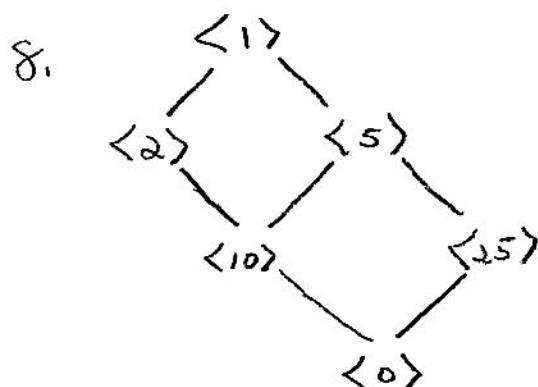
(c) $5^{-1} = 7$

(a) $U(8)$ is not a subgroup of $\mathbb{Z}_8$ because the operations are different.

(b) The set of nonnegative integers will not have additive inverses.

7. (a) $|D_{11}| = 22$ (b) elements have order 1, 2 or 11.

(c) 1 element of order 1, 11 elements of order 2, 10 elements of order 11.



9. $U(7) = \{1, 2, 3, 4, 5, 6\}$

has 3 as a generator since

$\langle 3 \rangle = \{3, 2, 6, 4, 5, 1\}$ (5 is also a generator of $U(7)$)

10. H is a subgroup of G , $x, y \in G$. Define $x \sim y \Leftrightarrow xy^{-1} \in H$.

Prove $\sim$ is an equivalence relation on G .

Reflexive $x \sim x$ for all $x \in G$ since $x \cdot x^{-1} = e \in H$.

Symmetric if $x \sim y$ then $xy^{-1} \in H$. Therefore $y \sim x$ because
 $yx^{-1} = (xy^{-1})^{-1} \in H$

Transitive if $x \sim y$ and $y \sim z$ we know xy^{-1} and yz^{-1} are in H .

Therefore $x \sim z$ since $xz^{-1} = (xy^{-1})(yz^{-1}) \in H$ by closure under the operation.

11. Prove every cyclic group must be abelian.

Let G be cyclic with $G = \langle a \rangle$.

Let x, y be any two elements of G . [show $xy = yx$]

Since $G = \langle a \rangle$ we know $x = a^j$ and $y = a^k$ for some integers $j \in \mathbb{Z}$.

Therefore $xy = a^j \cdot a^k = a^{j+k} = a^{k+j} = a^k \cdot a^j = yx$

12. Given G is Abelian and $H = \{x \in G : x^3 = e\}$. Prove H is a subgroup of G .

① $e \in H$ since $e^3 = e$ so H is not empty.

② Let $x, y \in H$ [show xy is in H]

$x^3 = e$ and $y^3 = e$. $(xy)^3 = xyxyxy = (xxx)(yyy) = x^3 \cdot y^3 = e \cdot e = e$

So $xy \in H$. H is closed under multiplication

③ Let $x \in H$. [show $x^{-1} \in H$]

$x^3 = e$ means $x \cdot x \cdot x = e$. Multiply both sides by x^{-1} three times to get
 $e = (x^{-1})^3$ so $x^{-1} \in H$.